NewOrbit Space Technical Assignment

Assignment solutions · Summary of previous experience · Plan for the first 1, 3 and 6 months

Om Anant Kulkarni
M.Sc. Aerospace Engineering, Technical University of Munich

31 October 2026

Executive technical summary

The five assignment tasks are addressed using beam energy and momentum balances, a finite-turn coil model, a literature-based assessment of propellant switching, real-gas storage calculations and a proposed verification programme. Calculated performance is conditional on the definition of measured currents and on transfer of empirical thrust correction factors. Where parameters are not specified, engineering assumptions are stated and their influence on the results is evaluated.

At 3 sccm (273.15 K, 101325 Pa) and 2000 V net acceleration, I interpret the screen reading as total extracted current and the accelerator current as primary-ion interception for the baseline calculation. This gives outgoing currents of 99 and 60 mA. The ideal singly charged beam estimates are 7.30 mN and 2543 s for xenon in Case 1, and 4.43 mN and 1541 s in Case 2. For equal extracted fractions of molecular nitrogen and oxygen ions, the corresponding estimates are 3.49 mN and 5316 s, and 2.12 mN and 3222 s. Applying literature-based effective correction factors separately gives approximately 7.08/4.29 mN for xenon and 3.00/1.82 mN for the mixture. These corrected values are extrapolations. I also evaluate an alternative current interpretation because the supplied readings do not uniquely establish the escaping momentum flux.

A 380-turn, air-core winding with a nominal 90 mm clear bore produces 5 mT at its centre at 1.125 A. Its calculated copper mass is 0.701 kg and its cold dissipation is 4.00 W. The field falls to 4.20 mT at the ends of the assumed 40 mm chamber. Therefore, this design meets a centre-field requirement; compliance with a minimum field throughout the volume requires an off-axis field map and assessment of the permitted peak field.

Two separate 1.1 litre tanks (2.2 L combined internal volume) require nominal usable inventories of 702.9 g xenon and 321.3 g mixture. With a proposed 10% operational allowance and a 2 bar absolute residual at 20 °C, loaded inventories become 785.2 g and 356.1 g. CoolProp HEOS calculations give initial pressures of 59.60 and 273.70 bar. The mixture reaches 317.83 bar at 50 °C. A conditional 300 bar design trade (not an assignment requirement) would require reducing the assumed 10% allowance; a limited load of 340.2 g nevertheless supports approximately 525 nominal hours, exceeding the requested 500 h, with approximately 5% usable-propellant margin.

The proposed verification programme covers unit and integrated testing, flight acceptance and life demonstration. Test criteria and outstanding design inputs are identified explicitly.

Contents

1 Requirements and analysis basis **3**

2 Task 1 Beam performance and current interpretation **4**

3 Task 2 Auxiliary magnetic system **7**

4 Task 3 Propellant transition and relevant research **9**

5 Task 4 Gas storage and feed system **11**

6 Task 5 Qualification and flight acceptance **13**

7 Design confirmations before implementation **24**

8 Conclusions **24**

One-page summary of my previous experience **26**

Plan for the first 1, 3, and 6 months **27**

1 Requirements and analysis basis

1.1 Assignment traceability

ID	Assignment requirement	Response
R1	Estimate thrust and specific impulse for both supplied current cases, on xenon and a 50:50 volume mixture at 3 sccm; state current, voltage, species and reference-state assumptions.	Section 2
R2	Design an adjustable 0.5–5 mT auxiliary axial magnetic system around a 90 mm inner diameter; specify winding and current.	Section 3
R3	Explain xenon-to-mixture consequences for thruster, cathode, performance, materials and lifetime.	Section 4
R4	Size separate 1.1 litre tanks for 500 h at 4 sccm xenon followed by 500 h at 8 sccm mixture.	Section 5
R5	Define unit and system qualification and flight acceptance, objectives, order, criteria, standards and flow diagrams.	Section 6

1.2 Assumptions and evidence classification

Assignment values are requirements (R). Calculated results are analysis (A). Literature measurements are evidence for their stated configuration (E). Proposed design or test choices are proposals (P), and unavailable hardware information is an open interface (O). These distinctions apply throughout.

Item	Basis and consequence	Status
Flow reference	273.15 K and 101325 Pa; 50:50 volume fraction treated as mole fraction at the reference state. Actual MFC reference and calibration remain to be confirmed.	P/O
Potentials	Screen +2000 V, accelerator -300 V relative to downstream reference. Source plasma offset and plume potential neglected in the nominal calculation.	P
Species model	Xe reference mass 131.293 u; mixture reference mass 30.0061 u. Empirical factors 0.97 and 0.86 are conditional extrapolations.	E/P
Magnetic geometry	40 mm axial chamber length adopted for the coil design. Turn positions, bore and winding build are specified explicitly.	P/O
Storage	20 °C fill state; -10 to +50 °C sensitivity envelope; 10% usable allowance; residual at 2 bar and 20 °C.	P
Neutraliser demand	Task 1 reports thruster-only specific impulse. Task 4 follows specified flows; separate cathode flow must be added if excluded from them.	O
Pressure limit	300 bar evaluated as a conditional design trade only, not an assignment requirement; certification, temperature dependence and oxygen-service approval not established.	O
Verification	Uncrewed spacecraft assumed. Launcher, local mounting spectra, thermal predictions and starts are not specified.	P/O

1.3 Constants and notation

$e = 1.602176634 \times 10^{-19}$ C, $N_A = 6.02214076 \times 10^{23}$ mol^{-1}, $R = 8.314462618$ J mol^{-1} K^{-1} and $g_0 = 9.80665$ m s^{-2}. I_s, I_a, I_b denote indicated screen current, accelerator current magnitude and outgoing beam charge current. F is axial thrust; $\dot{m}$ is supplied mass flow. V_n is net acceleration potential and V_t is screen-to-accelerator extraction potential. All pressures in this report are absolute unless specifically labelled otherwise.

2 Task 1 Beam performance and current interpretation

The specified volumetric feed is converted to mass flow, and ion momentum is calculated from the net acceleration voltage. Two current interpretations are evaluated because accelerator collection changes the escaping beam only under a defined current balance. Ideal estimates provide the baseline; empirical corrections are assessed separately.

2.1 Operating conditions and flow conversion

Table 2: Operating conditions used for the assignment calculations.

Parameter	Case 1	Case 2
Screen current	100 mA	100 mA
Screen voltage	2000 V	2000 V
Accelerator current magnitude	1 mA	40 mA
Accelerator voltage magnitude (given)	300 V	300 V
Accelerator potential (assumed)	-300 V	-300 V

For standard volumetric flow $\dot{V}_{std}$,

$$\dot{n} = \frac{p_{std}\dot{V}_{std}}{RT_{std}}, \qquad \dot{m} = M\dot{n}. \tag{1}$$

One sccm corresponds to 7.4358×10^{-7} mol/s under the selected reference. At 3 sccm, $\dot{n} = 2.2307 \times 10^{-6}$ mol/s, yielding 0.29288 mg/s xenon or 0.066936 mg/s mixture. The singly charged molecular-reference current at complete utilisation is $eN_A\dot{n} \simeq 0.2152$ A. For a dissociating or multiply charged mixture, this reference current is not a universal ceiling: atom and charge balances must replace the one-ion-per-feed-molecule assumption.

Note on the given currents: the screen current is 100 mA in both cases as supplied. Any value of 60 mA is a derived quantity under a particular current-loss interpretation (Model B below), not a given operating condition.

2.2 Energy and thrust equations

An ion born near potential V_p and escaping into a plume at V_∞ gains $ze(V_p - V_\infty)$. The nominal $V_n = 2000$ V therefore differs from the extraction potential $V_t = 2300$ V. The downstream ion does not retain the full accelerator-gap energy after leaving the negative accelerator potential. Source and plume potential corrections can be added when measured. The assignment specifies an accelerator-voltage *magnitude* of 300 V and explicitly requests a polarity assumption. I assume $V_a = -300$ V relative to the downstream plume, consistent with the conventional negative accelerator bias used to suppress electron backstreaming [2]. This does not change the supplied

magnitude. If +300 V were instead intended as the signed potential, the screen-to-accelerator difference would be 1700 V and the 40 mA primary-ion impact-power scale would be 68 W rather than 92 W. The ideal far-field ion energy would still be set by the 2000 V source-to-plume difference, provided ions escape and the assumed current remains valid; extraction, focusing and electron suppression would require reassessment. The reported thrust is therefore not calculated using either 2300 V or 1700 V. For a single species,

$$v_i = \sqrt{\frac{2zeV_n}{m_i}}, \quad F = \gamma I_b \sqrt{\frac{2m_i V_n}{ze}}, \quad I_{sp} = \frac{F}{\dot{m} g_0}. \tag{2}$$

These relations follow from energy and momentum conservation; conventional ion-thruster modelling is described in [2]. A species-resolved expression is

$$F_z = \sum_j I_j f_j \sqrt{\frac{2m_j V_{n,j}}{z_j e}}, \tag{3}$$

where I_j is charge current and f_j represents its axial momentum projection. An arithmetic average of species masses generally does not reproduce this sum. Number fractions and charge-current fractions must not be interchanged. As a separate idealised species sensitivity, at fixed singly charged beam current, voltage and divergence, replacing a 30.01 u molecular reference by a 15.00 u atomic reference gives $F_{atomic}/F_{molecular} = 1/\sqrt{2} = 0.707$, a 29.3% reduction. This comparison holds extracted current fixed, not particle throughput or ionisation efficiency; it is not an additional correction to the empirical mixture factor used below.

For an ideal collimated, singly charged beam, the molecular-mixture baseline uses equal extracted charge-current fractions of N_2^+ and O_2^+. Thus

$$F_{mix,ideal} = \frac{I_b}{2}\left[\sqrt{\frac{2M_{N_2}V_n}{N_A e}} + \sqrt{\frac{2M_{O_2}V_n}{N_A e}}\right]. \tag{4}$$

Equal extracted fractions are a modelling assumption, not a consequence of equal feed fractions. The ideal values below precede any empirical correction. The mean-mass approximation is approximately 0.055% higher for this molecular mixture.

Table 3: Ideal beam estimates at 3 sccm and 2000 V, with no divergence or empirical correction.

Beam current (mA)	Xe thrust (mN)	Xe I_{sp} (s)	Mixture thrust (mN)	Mixture I_{sp} (s)
100	7.378	2569	3.525	5370
99	7.304	2543	3.490	5316
60	4.427	1541	2.115	3222

Schwertheim et al. define their atmospheric correction relative to a 30 u molecular reference; composition and divergence are combined and were not independently resolved [1]. Consequently the present nominal mixture calculation uses that reference with $\gamma = 0.86$ once. A second fully dissociated-mass correction is not applied. The xenon value is $\gamma = 0.97$. The reported atmospheric factor is 0.86 ± 0.05, corresponding to about 5.8% relative uncertainty in the factor alone. This does not include the additional model uncertainty of transferring it to the assignment operating points. The published atmospheric screen voltage is 2000 V, with a −400 V accelerator; xenon mapping uses 1800 V and −400 V. The atmospheric 3 sccm assignment point lies below the paper's reported performance-map flow range of 5.8–14 sccm; the factors are extrapolations, not guaranteed assignment-point performance.

2.3 Current accounting and numerical answer

For the main estimate, the screen reading is taken as total extracted primary-ion current, and accelerator current is treated as intercepted primary ions: Model B below. If the reading instead denotes current collected by the screen-grid metal, the outgoing beam current cannot be inferred from these two readings alone. I use Model B to give a definite calculation under a stated current balance, not because direct interception has been established experimentally. Model A is retained as a sensitivity case for a different current definition or substantial momentum retained by fast neutrals:

1. Model A: I_s represents the thrust-producing extracted current sufficiently well, so use 100 mA in both cases.
2. Model B: I_s is total extracted primary-ion current and I_a is entirely direct primary interception. With negligible other paths, $I_b = I_s - I_a$, giving 99 and 60 mA.

Table 4: Conditional performance at 3 sccm and 2000 V.

Current model	Xe F (mN)	Xe I_{sp} (s)	Mixture F (mN)	Mixture I_{sp} (s)
A, either case: 100 mA	7.156	2492	3.033	4621
B, Case 1: 99 mA	7.085	2467	3.003	4575
B, Case 2: 60 mA	4.294	1495	1.820	2773

The Case 2/Case 1 ratio in Model B is $60/99 = 0.6061$, a 39.4% reduction only if the same effective correction factor is retained. Model A predicts no change from current accounting alone. Neither model proves the actual change in divergence. At equal current and voltage, the selected factors imply $F_{Xe}/F_{mix} = 2.359$. At equal standard flow and current, $I_{sp,mix}/I_{sp,Xe} = 1.855$. The often quoted $\sqrt{131.293/30.0061} = 2.092$ comparison assumes equal correction factors and cannot be substituted into this table's conclusions.

2.4 Accelerator current, perveance and erosion

A 40% accelerator-to-screen-current ratio is a strong diagnostic concern. Direct interception, charge exchange, secondary electrons, electron backstreaming and parasitic leakage must be distinguished using a circuit audit and measurements. Charge exchange can produce a fast neutral that retains much of the parent ion's momentum even though measured ion charge changes. A downstream charge-current measurement alone therefore need not capture total thrust. The supply-current definitions therefore remain essential to interpreting the result.

The planar Child–Langmuir relation illustrates the dependence

$$J_{CL} = \frac{4\epsilon_0}{9}\sqrt{\frac{2ze}{m_i}}\frac{V_t^{3/2}}{d^2}. \tag{5}$$

Actual multi-aperture optics require aperture geometry, plasma meniscus, hot gap, transparency and focusing. The voltage ratio $2000/2300 = 0.870$ alone diagnoses neither over-perveance nor crossover. The proposed diagnostic is an extraction-voltage sweep at controlled discharge conditions, recording screen/accelerator currents, direct thrust, angular ion current, grid temperatures, background pressure and discharge stability. Do not operate indefinitely at a suspected interception point merely to obtain a thrust value.

For an impacting species, the sputtered mass rate is

$$\dot{m}_{er} = \sum_j \frac{I_{a,j}}{z_j e} Y_j(E, \theta, T, \text{surface})\frac{M_{target}}{N_A}. \tag{6}$$

Collected current must first be corrected for non-ion contributions. Local recession also requires the spatial impact footprint; global mass loss cannot establish aperture life. A numerical erosion-rate ratio is therefore not assigned from accelerator-current ratio alone. Sputter yields must be selected for the actual species, impact energy, angle and grid surface before predicting recession.

A singly charged primary ion originating near +2000 V can reach the -300 V grid with approximately 2300 eV. If all 40 mA were such ions, incident kinetic power would be 92 W. The accelerator supply's $|V_a|I_a = 12$ W is a different quantity. Slow downstream ions falling from near zero potential would instead arrive near 300 eV. Actual heating also includes electron energy, recombination, radiation and conduction. These examples establish the need for a particle and electrical energy balance, not a temperature prediction.

Energy-conservation qualification: the high-voltage screen supply provides the ion acceleration energy. At 100 mA and 2000 V, the associated power scale is approximately 200 W. Therefore 92 W of grid impact power exceeding the RF discharge input does not violate energy conservation and cannot be used to rule out direct interception. It represents a serious thermal and lifetime concern, not a contradiction of the power budget. The NewOrbit experiments excluded operating points where accelerator current exceeded 4% of screen current and explicitly associated steep rises in accelerator current with ion-beam impingement [1]; the assignment's Case 2 is 40%, ten times that threshold. This does not prove the assignment device behaves identically, but it cautions against describing 40% accelerator current as normal, healthy charge-exchange operation.

2.5 Uncertainty and interpretation limits

For independent small uncertainties in a single-species reduced model,

$$\left(\frac{u_F}{F}\right)^2 \simeq \left(\frac{u_\gamma}{\gamma}\right)^2 + \left(\frac{u_I}{I}\right)^2 + \frac{1}{4}\left(\frac{u_V}{V}\right)^2. \tag{7}$$

Reference-state and calibration uncertainty enters $\dot{m}$ and hence I_{sp} and storage demand. It does not directly rescale thrust at fixed measured current, voltage and correction factor. Extrapolation and current-topology uncertainty are systematic model uncertainties and should be reported separately from instrument uncertainty. System specific impulse must include neutraliser consumption if supplied from the spacecraft.

3 Task 2 Auxiliary magnetic system

The chamber is short relative to its diameter, so I size the winding using the field of individual circular turns. This keeps the finite length, radial build and end-field reduction explicit. I then check resistance, dissipation and thermal sensitivity to assess whether the magnetic solution is electrically practical.

3.1 Topology and selected winding

I select an air-core solenoid and interpret the specified 0.5–5 mT range as the axial field at the discharge-chamber centre. The calculated axial profile is also reported to quantify variation over the chamber length.

The supplied illustration shows axial arrows directed towards a central plane and radial arrows directed away from the axis, consistent with a cusp-like field and an axial field reversal. A single solenoid does not reproduce that distribution. This is my interpretation of the written axial-field request; the assignment does not explicitly identify the field reference location. The design adopts

the illustrated 40 mm axial extent and 90 mm diameter, and treats the illustration as geometry and topology context. If reproducing its field reversal is mandatory, an opposing-current coil arrangement and a specified off-centre field reference are required; a symmetric cusp has zero field at its centre and cannot satisfy a nonzero centre-field requirement. I retain the solenoid as the preliminary design under this stated interpretation.

The proposed coil uses 38 turns per layer and ten layers of 0.912 mm bare copper diameter, with an assumed finished diameter of 1.032 mm. Turn-centre radii are $r_k = 45 + (k + 1/2)1.032$ mm for $k = 0, \ldots, 9$, and axial turn centres are $z_l = (l - 18.5)1.032$ mm. This is a 39.216 mm winding length and a 10.32 mm radial build. The 90 mm bore is measured to the inner wire surface; a former, chamber clearance, insulation system and tolerances increase the required radius and must be incorporated before manufacture.

Dimensional distinctions: the following quantities are not automatically equal and must be kept distinct: clear bore diameter, thruster outside diameter, winding inside diameter, bobbin thickness, assembly clearance, winding outside diameter, and axial winding envelope. The present model places the inner surface of the insulated winding at 45 mm radius with no separately modelled bobbin wall or assembly clearance. If a bobbin or clearance is introduced, the turn radii and all derived quantities must be recomputed. This is primarily a mechanical-traceability issue, not an error in the Biot–Savart implementation.

$$B_z(z) = \sum_{k,l} \frac{\mu_0 I r_k^2}{2[r_k^2 + (z - z_l)^2]^{3/2}}. \tag{8}$$

This finite-turn calculation is numerical evaluation of Biot–Savart, not FEM. At the centre it gives $I = 1.125$ A for 5 mT, or 0.1125 A for 0.5 mT. Total ampere-turns at maximum field are 427.5 A-turns. The end field at $z = \pm 20$ mm is 4.195 mT, 16.1% below the centre. Off-axis field and integration with any existing magnetic circuit require a volume map; centre compliance is not volume compliance.

Parameter	Calculated value
Copper length	119.76 m
Copper area	0.6533 mm^2
Resistance at 20 °C	3.161 Ω
Maximum current density	1.72 A/mm^2
Cold dissipation	4.001 W
Cold terminal voltage	3.56 V
Copper mass	0.701 kg

3.2 Thermal and electrical integration

For current control,

$$I^2 R_{20}[1 + \alpha(T - T_0)] + Q_{in} = G(T - T_{mount}) + \epsilon\sigma A(T^4 - T_{sur}^4). \tag{9}$$

The radiation-only screening case uses $G = 0$, $Q_{in} = 0$, $A = 0.0204$ m^2, $T_{sur} = 293.15$ K and $\alpha = 0.00393$ K^{-1}. The solved temperatures are 117.7 °C for $\epsilon = 0.3$ and 60.5 °C for $\epsilon = 0.8$. These are assumed-environment cases, not operational bounds when plasma or structure heating is present. Effective area, view factors, mounting conductance and surface finish require verification.

I propose a current-regulated supply with 6 V compliance as a preliminary allowance for hot winding resistance and harness losses. Include current telemetry, overtemperature shutdown, controlled ramping and an inductive-current discharge path sized after inductance is measured or calculated for the final assembly. Insulation and potting require qualified temperature, vacuum and

dielectric performance. Avoid a continuous conducting former that creates strong RF eddy-current coupling; verify any split former mechanically and thermally. Coil location, cabling and RF shielding must be tested for changes in discharge impedance and electromagnetic compatibility.

3.3 Field uniformity and design trade

The selected coil meets the centre-field interpretation of the requirement. If the on-axis field over the 40 mm chamber length must remain above 0.5 mT, its lowest commanded current must increase to approximately 0.134 A. If every point must reach 5 mT simultaneously, a centre-peaked winding is unsuitable when 5 mT is also a strict upper limit. A longer winding or a distributed coaxial coil arrangement should then be optimised for the specified volume and uniformity. The 0.134 A result covers the axis only; an off-axis map is needed before claiming compliance throughout the chamber volume.

At fixed geometry, copper mass and loss trade approximately as $Pm = \rho_e \rho_{Cu} (NI)^2 \ell_{turn}^2$. Changing wire size also changes winding geometry and thermal conditions, so this is a trade relation rather than an invariant guaranteeing identical performance. The selected ground-design winding is intentionally conservative in copper mass; a flight allocation must be agreed before optimising it.

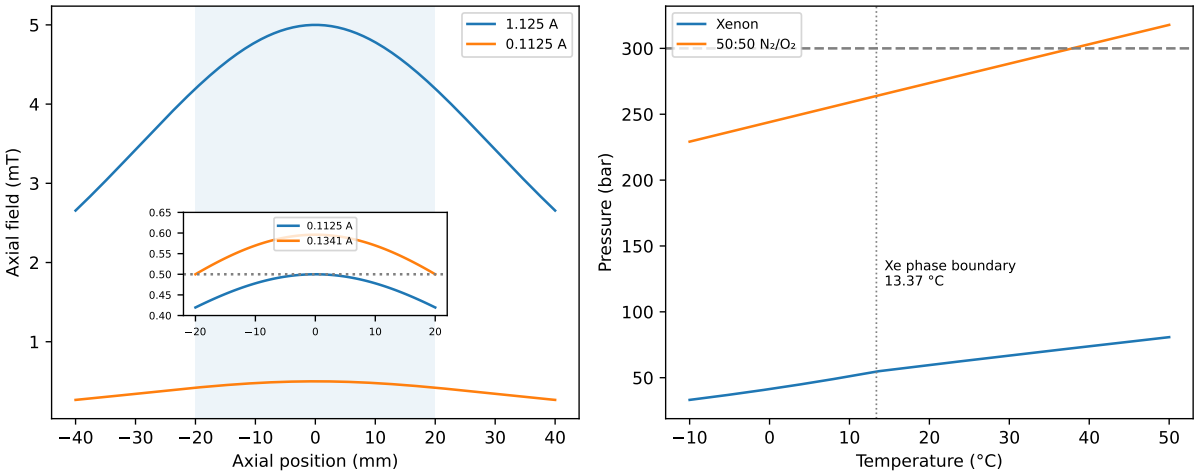


Figure 1: Calculated results. Left: on-axis coil field; the inset resolves the lower-current settings over the chamber length. Right: equilibrium tank pressure at fixed loaded mass and volume. At the selected xenon inventory, condensation begins at approximately 13.37 °C on cooling; below this boundary pressure follows saturation while phase fraction changes. Pressure is inventory-independent only while both phases remain present. The 300 bar line is a conditional interface limit.

4 Task 3 Propellant transition and relevant research

The change of propellant is assessed through three coupled effects: discharge energy losses, extracted ion composition and surface chemistry. These determine the useful performance comparison, the material candidates and the measurements needed to establish life.

4.1 Discharge and extraction

Replacing xenon with nitrogen and oxygen changes the particle balance, energy balance and RF load. A useful modelling basis is a species-resolved global balance with reaction coefficients

$$k_r = \int_0^\infty \sigma_r(E) v(E) f_E(E)\, dE, \qquad P_r = V_{pl} n_e n_s k_r \Delta E_r. \tag{10}$$

An EEDF consistent with the discharge must be supplied; assuming a Maxwellian distribution is itself a modelling choice. Molecular excitation, dissociation, electronic excitation, ionisation, attachment and detachment compete with wall losses. Rotational and vibrational losses cannot be ranked solely by threshold energy. The RF-air model of Lopez-Uricoechea et al. provides a relevant global-model framework, with its approximations restricted to its operating regime [3].

Composition changes the Bohm flux, space-charge extraction, effective mass-to-charge ratio and beamlet focusing. An optics set optimised for one gas cannot be assumed optimal for another. Electron attachment in oxygen can alter electron density and sheath behaviour, but electronegativity must be measured or calculated rather than assumed large. Changed conductivity affects skin depth, matching and absorbed power. The engineering response is independent ignition and operating maps for both gases, including stable transitions and loss-of-flow recovery.

4.2 Performance comparison and neutralisation

AURA is directly relevant because it reports an RF gridded engine and RF neutraliser on both propellants; its gas configurations use different ion optics and cathode aperture. Its demonstrated configuration change is not evidence of unrestricted switching with identical hardware [1]. For this assignment, either a common optics/cathode configuration must be qualified for both gases or the required configuration change must be declared.

At the assignment's fixed current and voltage, the adopted model predicts mixture thrust to be 42.4% of xenon thrust, while thruster-only specific impulse at equal sccm is 1.855 times higher (Section 2). These are momentum and mass-flow comparisons, not predictions of equal-power operation. Molecular energy losses and neutraliser overhead can reduce the apparent system-level benefit.

A quantitative discharge-cost penalty requires paired power and beam-current measurements at stated operating points. A meaningful comparison requires the same definition of input power, current utilisation, beam voltage and operating constraint. Record thruster RF input, absorbed plasma power where measurable, grid-supply power, cathode power, PPU losses and auxiliary power separately. Compare total system efficiency as $F^2/(2\dot{m}_{total}P_{total})$ only with consistent system boundaries.

Neutraliser performance must demonstrate electron delivery over the beam-current range, coupling potential, ignition at hot/cold conditions, and fault recovery. An RF plasma neutraliser is a justified candidate for oxygen-bearing operation; its power, gas consumption and thermal load must be included. The minimum electron-current capability is set by the full operating map and transient requirement, not only the nominal 100 mA assignment point. Cathode availability and stable current coupling are system release criteria.

4.3 Materials and life mechanisms

Grid and cathode candidate selection must combine ion sputtering and chemical exposure. A material that forms an oxide is not automatically self-protecting under energetic bombardment. Similarly, oxide volatility is not enough to reject a material without actual temperature and flux.

Candidate refractory metals, coatings or carbon-based materials require comparative coupons at representative oxygen species, ion energy, temperature and duty cycle. Assess dimensional recession, coating adhesion, electrical conductivity, contamination and redeposition. High-pressure oxygen compatibility is a separate assessment from low-pressure plasma resistance.

For a provisional materials baseline, select molybdenum for the RF-neutraliser orifice and retain titanium as a comparative test article. Zorn et al. report greater central oxidation of titanium than molybdenum orifices after four hours in oxygen; this supports screening priority, not a 500 h life claim [9]. Their titanium grids behaved differently from the neutraliser, illustrating the importance of local exposure.

Candidate	Proposed use and selection condition
Molybdenum	Neutraliser-orifice baseline; grid candidate. Verify oxidation, recession and emission/coupling stability at the actual temperature and ion flux.
Titanium	Grid and orifice comparison candidate; retain only if dimensional and electrical stability meet the exposure requirement.
Tungsten	Refractory-metal screening alternative; assess fabrication, mounting and oxygen-exposure behaviour before selection.
Carbon-based materials	Do not select solely for low physical sputtering: oxygen-driven chemical erosion requires separate evidence.

No material ranking is transferred to the high-pressure oxygen-bearing feed system. Final down-selection combines representative plasma exposure with manufacture, thermal stress and contamination constraints.

Proposed diagnostics are a Faraday angular scan, an energy analyser, mass-to-charge-resolved sampling and direct thrust. Account for instrument transmission, angular bias and overlapping mass-to-charge signatures; an on-axis composition measurement alone is not the integrated beam composition. Optical emission supports discharge-state monitoring but does not by itself quantify extracted species fractions.

A representative programme should map pressure sensitivity with the same thruster settings at multiple facility backgrounds, recording gas-corrected pressure, pumping speed and wall conditions. Separate genuine propellant effects from facility charge exchange and plume return flux. Lifetime demonstration tracks grid apertures and gaps, neutraliser surfaces, RF insulation, matching drift, contamination, valves, regulators and PPU component temperatures. Coupon tests support material down-selection; representative endurance testing addresses operational life.

5 Task 4 Gas storage and feed system

The assignment specifies two separate 1.1 litre tanks: one for xenon and one for the 50/50 N_2/O_2 mixture. The combined nominal internal tank volume is therefore 2.2 L. Each tank is sized for its own 500 h segment; the total scheduled thruster operation is 1000 h. Each specified flow is integrated over 500 hours before adding an explicit usable allowance and residual. Pressure is then evaluated at the resulting tank density using real-gas properties because the required densities make an ideal-gas design assumption unsuitable, particularly for xenon.

5.1 Inventory and real-gas calculation

At the adopted standard reference, required usable mass is

$$m_u = q\left(\frac{p_{std}10^{-6}}{60RT_{std}}\right)tM, \quad m_L = (1+f)m_u + m_r, \quad m_r = V\rho(p_r, T_r). \tag{11}$$

Here q is in sccm, $t = 500 \times 3600$ s, $f = 0.10$, $p_r = 2$ bar, $T_r = 293.15$ K and $V = 0.0011$ m^3. The residual is an engineering assumption requiring regulator inlet-margin validation.

Table 5: HEOS storage calculation, CoolProp 8.0.0.

Quantity	Xenon	50:50 mixture
Flow (sccm)	4	8
Usable mass (g)	702.917	321.294
Residual mass (g)	11.983	2.711
Loaded mass (g)	785.191	356.134
Fill density (kg/m^3)	713.810	323.758
Pressure at -10 °C (bar)	33.140	229.192
Pressure at 20 °C (bar)	59.604	273.695
Pressure at 30 °C (bar)	66.802	288.446
Pressure at 40 °C (bar)	73.842	303.159
Pressure at 50 °C (bar)	80.777	317.834

The script uses the pure-xenon HEOS fluid and an explicit equimolar Nitrogen&Oxygen HEOS mixture with the library's built-in binary parameters; it does not use the predefined terrestrial-air mixture [4]. The gas phase is imposed for the mixture density–temperature evaluation in this warm temperature interval. Xenon uses equilibrium evaluation without that imposed phase. Model uncertainty, tank volume tolerance and composition tolerance are not represented by the extra digits, which are provided for reproducibility. The tabulated values have been independently reproduced by re-running the script with CoolProp 8.0.0 on a separate installation; a Peng–Robinson cross-check for xenon agrees within approximately 1%, consistent with the expected difference between a cubic EOS and the multiparameter HEOS backend.

The ideal-gas expression $p = mRT/(MV)$ is a useful check but substantially overestimates the xenon pressure near this fill state. The HEOS result is used for the design inventory, with pressure sensitivity evaluated over the selected temperature range.

5.2 Xenon phase state

At a fixed fill density, coexistence requires

$$\rho_{v,sat}(T) < \rho_L < \rho_{l,sat}(T). \tag{12}$$

Solving $\rho_{v,sat}(T) = 713.810$ kg/m^3 gives an onset temperature of **13.369 °C**. Below that point along the initial-load isochore, the equilibrium state lies in the two-phase region over the cold interval evaluated. The critical temperature alone does not determine onset. Consumption changes density and therefore changes the crossing temperature.

A controlled warm operating range can simplify withdrawal, but heater power, loss of heating and cold survival must be analysed. In microgravity, outlet phase depends on surface tension, wetting, geometry and thermal history. A ground "gas at the top" assumption is invalid. Specify acceptable regulator inlet state and assess whether thermal control alone suffices or a qualified acquisition strategy is required. Pressure is a weak inventory estimator near saturation; use integrated calibrated flow with uncertainty and appropriate pressure/temperature consistency checks.

5.3 Pressure allowance and duration trade

At 50 °C, limiting the mixture to 300 bar yields $m_L = 340.206$ g with the same EOS. Subtracting the 2.711 g residual gives nominal endurance

$$t_{nom} = \frac{340.206 - 2.711}{321.294} 500 = 525.21\,\text{h}. \tag{13}$$

Retaining a full 10% operational allowance gives $t_{10\%} = 477.47$ h. Therefore the nominal 500 h requirement can be met under this conditional limit, with approximately 5.04% usable allowance. These two numbers answer different questions: 525 h is the duration achievable; 477 h is the duration achievable while also holding a 10% reserve.

Conditional trade qualification: a 300 bar limit is an optional design trade, not a requirement stated in the assignment. If such a limit were imposed, the following must be stated carefully: a 310 bar allowable pressure does not satisfy a predicted 317.8 bar at 50 °C; a 40 °C limit does not satisfy a strict 300 bar constraint when the predicted pressure is 303.2 bar (linear interpolation suggests approximately 37.8 °C as the threshold; the actual value requires EOS evaluation with margin); and altering the 50/50 composition is not a compliant remedy. Increasing the oxygen fraction does not automatically improve pressure capacity at fixed required standard volumetric consumption, since molar demand and EOS effects must both be considered. If the full load and 50 °C maximum are retained, the calculated operating-pressure requirement is approximately 318 bar absolute, subject to EOS verification and the project's additional uncertainty and qualification requirements. This is a calculated operating pressure, not a proof or burst rating.

A published vessel description is not a certificate establishing oxygen compatibility, maximum allowable pressure at every temperature or applicability to both assignment tanks. Obtain the certified rating, material, liner, proof history, temperature range and life limitations. Also distinguish qualification test temperature from maximum fully filled mission temperature; the relevant combined load case must be defined rather than assumed.

5.4 Feed architecture and oxygen compatibility

The proposed feed system has independent tank isolation and pressure reduction paths, pressure and temperature telemetry, filtration, flow control, and downstream selection with prevention of crossflow. A proposed chain is tank service/fill valve, isolation valve, filter, regulator, low-pressure sensor and metering device. The final arrangement must address trapped volumes, regulator failure, leak containment, purging and safe servicing. Verify regulator droop at end-of-life inlet pressure and minimum flow, and characterise oxygen/mixture calibration directly.

The 50% oxygen mixture requires a documented oxygen compatibility and cleanliness assessment. ASTM G93/G93M-25 is the applicable cleanliness guide to consider; it does not by itself select one universal cleanliness level [8]. Select materials and lubricants for the actual pressure, temperature, gas composition and ignition mechanisms. Control particles, hydrocarbons, rapid pressurisation, adiabatic compression and rubbing/impact ignition. Cleaning validation and cleanliness preservation through assembly are required.

Pressure proof and burst factors follow the selected hardware classification and maximum design pressure. Tanks, lines and valves require their applicable verification routes.

6 Task 5 Qualification and flight acceptance

Design qualification and flight-unit acceptance answer different questions. Qualification establishes environmental and life capability; acceptance checks the delivered hardware against that qualified

configuration. Unit tests establish subsystem performance before integrated tests address coupling, transitions and faults.

6.1 System boundary and adopted mission context

The test article is the complete ion propulsion string: thruster, RF neutraliser, PPU and RF generators, propellant management equipment, both tanks, flight-representative harnesses, feed lines, mounting interfaces and control software. Spacecraft power, command, thermal and structural interfaces are represented by validated simulators or representative hardware. Spacecraft-level verification remains a separate activity; it is not replaced by propulsion-string testing.

NewOrbit describes NEO-1 operation at 200–300 km over a five-year mission, an AURA total-impulse design target of 1 MN s, capability exceeding 30,000 on/off cycles, and more than 1000 cumulative hours in pure oxygen plasma without visible grid degradation [10]. I use these manufacturer statements as context for endurance, ignition cycling and oxygen exposure. They do not replace the assignment's 500 h xenon followed by 500 h mixture duty or establish qualification of the proposed configuration. Existing test evidence receives heritage credit only after configuration, exposure, diagnostics and failure criteria have been compared.

The NewOrbit publication also reports vibration testing at 14 grms using GSFC-STD-7000A qualification levels, with low-level sine sweeps before and after [1]. This is used as contextual project evidence, not as an automatic requirement for this assignment. Its scope excludes gas and electrical interfaces, which need separate verification. Published engineering-model testing is not equivalent to qualification of a complete proposed system.

I adopt dedicated qualification hardware and a representative endurance string, with acceptance testing on each flight unit before integration and a final integrated acceptance campaign. Qualification environments come from spacecraft interface loads and thermal analysis; orbital altitude alone does not define vibration spectra or equipment temperatures. The numerical planning choices below are my proposed baseline, subject to the cited standards and hardware limits.

6.2 Verification philosophy and standards

Qualification establishes design adequacy over an agreed environment and life. Flight acceptance checks the delivered unit's workmanship and conformance without unnecessarily consuming life. Protoflight is a separately approved model philosophy, not an automatic choice for PPU or PMS. Use dedicated qualification articles for destructive pressure tests and wear-sensitive endurance where practical. Flight use of an environmentally tested article requires life accounting and approved disposition.

Basis	Application
ECSS-E-ST-10-02C Rev.1 (2018) [5]	Verification control, evidence, requirement closure and review.
ECSS-E-ST-10-03C Rev.1 (2022) [6]	Equipment and element testing, including environmental exposure, functional checks, uncertainty and retesting. The applicable schedules are tailored to article level and mission environment.
ECSS-E-ST-32-02C Rev.2 (2025) [7]	Classification and structural verification of pressurised hardware. Minimum factors and verification route depend on category, construction and load.

Basis	Application
ASTM G93/G93M-25 [8]	Cleaning and cleanliness assessment for oxygen-enriched equipment; project cleanliness requirement remains to be specified.
Project specifications	EMC limits, electrical derating, contamination budget, launcher environment, spacecraft thermal interfaces and operational constraints must be baselined before test release.

The programme uses these governing standards with documented project tailoring. Numerical levels and equipment applicability are confirmed at test readiness review.

6.3 Levels and durations

For initial planning, I propose random-vibration qualification at the maximum expected PSD plus 3 dB for two minutes per axis, and acceptance at the agreed limit spectrum for one minute per axis. I propose shock qualification at the maximum expected SRS plus 3 dB, with the number of applications set by the life-cycle environment. Sine qualification uses K_Q times the limit-load spectrum, with K_Q obtained from the structural basis. These are provisional equipment-level planning values; the released procedure must identify the applicable schedule in ECSS-E-ST-10-03C Rev.1 and record any tailoring [6].

I propose eight qualification thermal cycles and four acceptance cycles as the initial equipment planning baseline, pending applicability review. Define predicted operating and non-operating limits, then acceptance margins and additional qualification margins separately. Dwell begins after monitored critical nodes stabilise; sufficient dwell is required for the complete functional test at each relevant extreme. High-voltage operation during pumpdown is inhibited unless specifically qualified for that pressure regime. Thermal balance and thermal cycling have different purposes and should have separate procedures.

A larger overall g_{rms} does not establish spectral coverage. Qualification similarity requires frequency-by-frequency PSD/SRS comparison, duration, axes, mounting stiffness, notching, response limits and configuration. Pressure tests use approved category-specific factors applied to the defined maximum design pressure, with controlled fluid, temperature, hold time, deformation/leak checks and safety provisions. Destructive burst testing is performed on dedicated articles. No universal proof/burst pair is assigned here without hardware classification.

6.4 Qualification and acceptance allocation

The matrix below defines how tests are allocated. Q denotes qualification hardware; F denotes flight hardware. Environmental testing uses flight-representative mounting and connections. For every test, the data package records article serial number, configuration, procedure revision, calibrated instrumentation, uncertainty, commanded and measured exposure, anomalies and signed disposition.

Test and article	Qualification exposure and duration	Flight acceptance and decision rule
Mechanical: units and integrated assembly	Q: proposed random PSD +3 dB, 2 min/axis; tailored sine and applicable shock. Integrated exposure verifies mounting, harness and feedline interactions not bounded by unit evidence.	F: proposed limit PSD, 1 min/axis where allocated. No automatic repetition at every level. Pass: no damage, prohibited resonance change, leak increase or functional nonconformance; response limits prevent overtest.
Thermal vacuum: units and propulsion string	Q: proposed 8 cycles at derived qualification limits; hot/cold starts and operation, survival exposure, separate thermal-balance cases.	F: proposed 4 cycles at acceptance limits where allocated. Pass: function and insulation retained; temperatures plus uncertainty remain within qualified component limits.
Pressure: tanks and pressure-bearing PMS	Q: classification-based proof, pressure cycles and dedicated burst evidence. Fill/drain cycles and temperature combinations represent service life.	F: required NDT, proof and external/seat leak checks under the selected verification route. Pass: no prohibited permanent deformation; leakage within allocated limits. No flight burst test.
Electrical and EMC: PPU/RFG and integrated string	Q: bus extremes, representative loads, insulation, grounding, emissions/susceptibility, arcs and fault responses. Exercise ignition and switching as well as steady firing.	F: calibrated outputs, insulation and protection checks; integrated command and telemetry test. Pass: allocated limits met; no unintended reset, hazardous output or command.
Propulsion operation: coupled thruster and neutraliser	Q: both-gas maps at minimum, nominal and maximum approved points, including thermal extremes and end-of-life feed pressure.	F: abbreviated reference-point firing on both gases. Pass: approved performance envelope and neutralisation maintained, no abnormal grid current or unstable discharge.
Endurance: representative propulsion string	Q/life: 500 h Xe at 4 sccm, then 500 h mixture at 8 sccm; checkpoints every 100 h and at transition. Additional allowance justified separately.	F: no mission-duration endurance exposure. Acceptance firing is limited to stabilisation and repeatable checks; consumed life is recorded.

Thermal levels and dwell. For each interface or controlled equipment node, I define predicted hot/cold limits T_{PH} and T_{PC} from the thermal model. My provisional temperature margins are 5 K for acceptance and a further 5 K for qualification:

$$T_{AH} = T_{PH} + 5\,\mathrm{K}, \quad T_{AC} = T_{PC} - 5\,\mathrm{K}, \quad T_{QH} = T_{AH} + 5\,\mathrm{K}, \quad T_{QC} = T_{AC} - 5\,\mathrm{K}. \tag{14}$$

These margins are planning assumptions, not claimed universal ECSS values. Operating and survival limits are treated separately and checked against component ratings. I propose stabilisation when monitored critical-node drift is below 1 K/h for one hour, followed by at least one hour of dwell or the complete functional sequence, whichever is longer. High-inertia hardware may need longer. Tank temperature and pressure are assessed together; a fully loaded mixture tank cannot simply be heated beyond its allowable pressure envelope to satisfy a generic thermal margin.

Pressure levels and hold times. For each pressure boundary, I establish the maximum design pressure from fill conditions, thermal excursions, regulator failure and trapped-volume cases. The released plan identifies its ECSS hardware category and specifies $p_{proof} = K_p p_{design}$ and, where applicable, $p_{burst} = K_b p_{design}$. Factors, hold times, pressurisation rates and test fluid follow that classification and the approved vessel or component procedure. The planning assumption is procurement of qualified vessels with traceable structural evidence; custom-vessel development would require an expanded programme. The 2 bar inventory residual is not a structural test pressure. A post-regulator line is assessed against its credible failure pressure or demonstrated relief/isolation protection.

Functional duration and fault response. I propose three consecutive five-minute steady-data windows at each selected reference point after thermal and discharge stabilisation. Qualification covers the approved operating envelope; flight acceptance uses the reduced reference-point set and records accumulated firing time. Fault tests cover loss of neutraliser current, grid arcs, feed interruption, bus undervoltage and sensor faults. Measured shutdown/recovery times are compared with limits derived from the grid energy, PPU protection and thermal damage budgets before fault injection. A safe shutdown alone does not qualify automatic recovery.

6.5 Unit test matrix

Unit	Qualification objective and tests	Flight acceptance and evidence
Thruster	Inspection and grid metrology; insulation; both-gas performance maps; ignition; thermal and vibration exposure; applicable shock; hot/cold operation; grid erosion and endurance.	Serialised dimensions; leak/electrical checks; abbreviated reference-point firing; environmental screening; post-test current, thrust and temperature compliance.
Neutraliser	Emission-current map, coupling voltage, ignition/restart, oxygen exposure, thermal environment, endurance and plume coupling.	Emission at required current; hot/cold start checks; verified consumption and temperature; stable coupled beam operation.
PPU/RFG	Input voltage extremes, regulation, efficiency, ripple, load transients, arc/recycle protection, fault injection, thermal derating, EMC and insulation.	Workmanship inspection; calibration; input/output and protection checks; thermal/environmental screening with representative loads.
PMS	Flow accuracy across inlet pressure, regulator droop, seat/external leakage, isolation/crossflow, cleanliness, thermal operation and cycle life.	Cleanliness certification; proof as applicable; leak and seat tests; calibrated flow at nominal and end-of-life inlet pressure.
Tank	Material/process records; structural and fracture assessment; proof; dedicated burst and cycle verification as applicable; leak, thermal and mounting-load compatibility.	Traceable certified vessel; required NDT/proof/leak evidence; dimensional, cleanliness and service-life records.

Each environmental stage is bracketed by inspection and reduced functional tests. For propulsion hardware, external leak checks and alignment-sensitive measurements follow mechanical exposure.

PPU and PMS life mechanisms include capacitor ageing, switching/thermal stress, insulation degradation, valve cycles, seal creep and pressure cycling; they are not exempt from life verification.

6.6 Integrated system test matrix

Test	Objective and measurements	Decision criterion
Magnetic-field mapping	Calibrated three-axis Hall probe; current sweep, centre and axial scans, and an off-axis grid at cold and hot steady states.	Centre range 0.5–5 mT within agreed uncertainty; compare measured profile with the finite-turn model. Volume limits are agreed separately.
Interface verification	Polarity, grounding, bonding, command/telemetry, pressure boundaries, flow routing and hardware configuration.	All interfaces match released drawings; no unexplained current or flow path.
Initial vacuum firing	Establish thrust, grid currents, neutraliser coupling, flow, total power and temperatures.	Meets agreed performance envelope with uncertainty accounted for.
Operating map	Both propellants, input voltage, flow extremes and relevant hot/cold states.	Stable ignition and firing; no unqualified interception, backstreaming or overheating.
Transition and faults	Gas switching/purge, loss of flow, neutraliser dropout, arcs, bus dips and safe shutdown.	Safe-state transition demonstrated; no damage; recovery within agreed operational limits.
Thermal vacuum	Representative mounting conductance and thermal boundary; operating and survival cases.	All temperatures within qualified limits; functional compliance before and after exposure.
EMC and spacecraft coupling	Conducted/radiated disturbances; command integrity; grounding and plume interfaces.	Compliance with allocated spacecraft limits, with no unintended reset or command.
Endurance evidence	Throughput, accumulated impulse, starts, degradation and configuration changes.	Mission allocation covered with approved life rationale and residual margin.

Test pumping must support representative background pressure. Log gas species, gauge correction, calibration status, facility configuration and thrust-balance tare. A short functional firing cannot substitute for an endurance campaign or reproduce all in-orbit plume interactions.

6.7 Measurable criteria and provenance

Assignment R1 requests estimates, not a contractual thrust minimum. The values calculated in Section 2 must not silently become acceptance specifications. The following quantitative proposals make test planning concrete while retaining their approval status.

Observable	Proposed decision rule	Basis
Thrust and I_{sp}	$F - U_F \geq F_{req}$ and $I_{sp} - U_{Isp} \geq I_{sp,req}$ at each required point. Establish absolute requirements before acceptance.	Derived rule; thresholds O
Baseline stability	Post/pre thrust change within 5% after combined measurement uncertainty; still meet absolute limits.	P; revise from budgets
Flow accuracy	Error plus expanded uncertainty no greater than 2% of reading over the specified working range.	P; calibration feasibility needed
External leakage	Initial planning limit 10^{-6} mbar litre/s helium per sealed unit; translate to propellant loss and mission budget before adoption.	P; not an ECSS universal limit
Accelerator current	Below the qualified gas-specific current-ratio envelope with no abnormal temperature or instability. I would establish the allowable ratio for each qualified operating envelope.	Derived; envelope O
Electrical insulation	No breakdown at the approved dielectric test voltage; insulation resistance above approved humidity/temperature-dependent requirement.	Derived; voltage and resistance O
Pressure integrity	No prohibited deformation or leakage after prescribed proof; structural and burst evidence accepted by category.	Standard/project
Thermal limits	Measured temperature plus uncertainty below qualified limit; survival cases pass subsequent functional test.	Derived
Ignition and recovery	All scheduled qualification sequences complete; cycle count and allowed ignition time from mission requirements.	Count/time O

For a maximum limit L, pass only when the measured value plus its agreed uncertainty guard band is $\leq L$. For a minimum requirement reverse the guard band. The test readiness review approves the remaining numerical limits before execution. Reduced functional tests use the same calibrated reference points to avoid conflating operating-point shifts with degradation.

Unit-specific functional criteria. For the neutraliser, I propose demonstrating at least 1.2 times the maximum approved beam-current demand as a development margin, with stable coupling within the qualified voltage envelope. This margin is an assumption, not a published AURA specification. Flight acceptance checks the allocated current plus operational margin without re-demonstrating destructive limits.

For PPU outputs, I propose DC regulation error plus uncertainty no greater than 1% of command over the approved bus/load range; ripple and transient limits are allocated from beam stability and insulation requirements. Oscilloscope bandwidth, probe loading and measurement locations are specified. RF forward/reflected power and matching stability are recorded separately. Arc response must satisfy the released energy and recovery-time limits.

For the PMS, the 2% delivered-flow criterion is verified across tank pressure, temperature and both gases, including the residual-pressure operating point. Outlet pressure must remain inside the metering device's qualified inlet range. Seat leakage is assessed separately from external leakage against unintended-flow and isolation requirements. Tank acceptance requires traceable

material/process records and completion of the selected pressure and cleanliness verification route.

For the thruster, pre/post reference-point thrust must satisfy the stated 5% stability rule with uncertainty guard bands, while absolute performance, accelerator current, grid geometry and thermal limits remain satisfied. Aperture and hot-gap tolerances are tied to the ion-optics drawing and backstreaming margin; visual inspection alone is insufficient.

6.8 Life strategy and qualification allocation

The specified duty is 500 h xenon followed by 500 h mixture. Its nominal thruster throughput is 0.7029 kg xenon and 0.3213 kg mixture. If one thruster must perform both segments, test that sequence or demonstrate why separate articles represent it; cumulative surface chemistry and switching history can matter. Determine cathode throughput independently. Total impulse must be integrated from the actual duty map, not calculated by transferring Task 1's 3 sccm performance to Task 4's 4/8 sccm operating points.

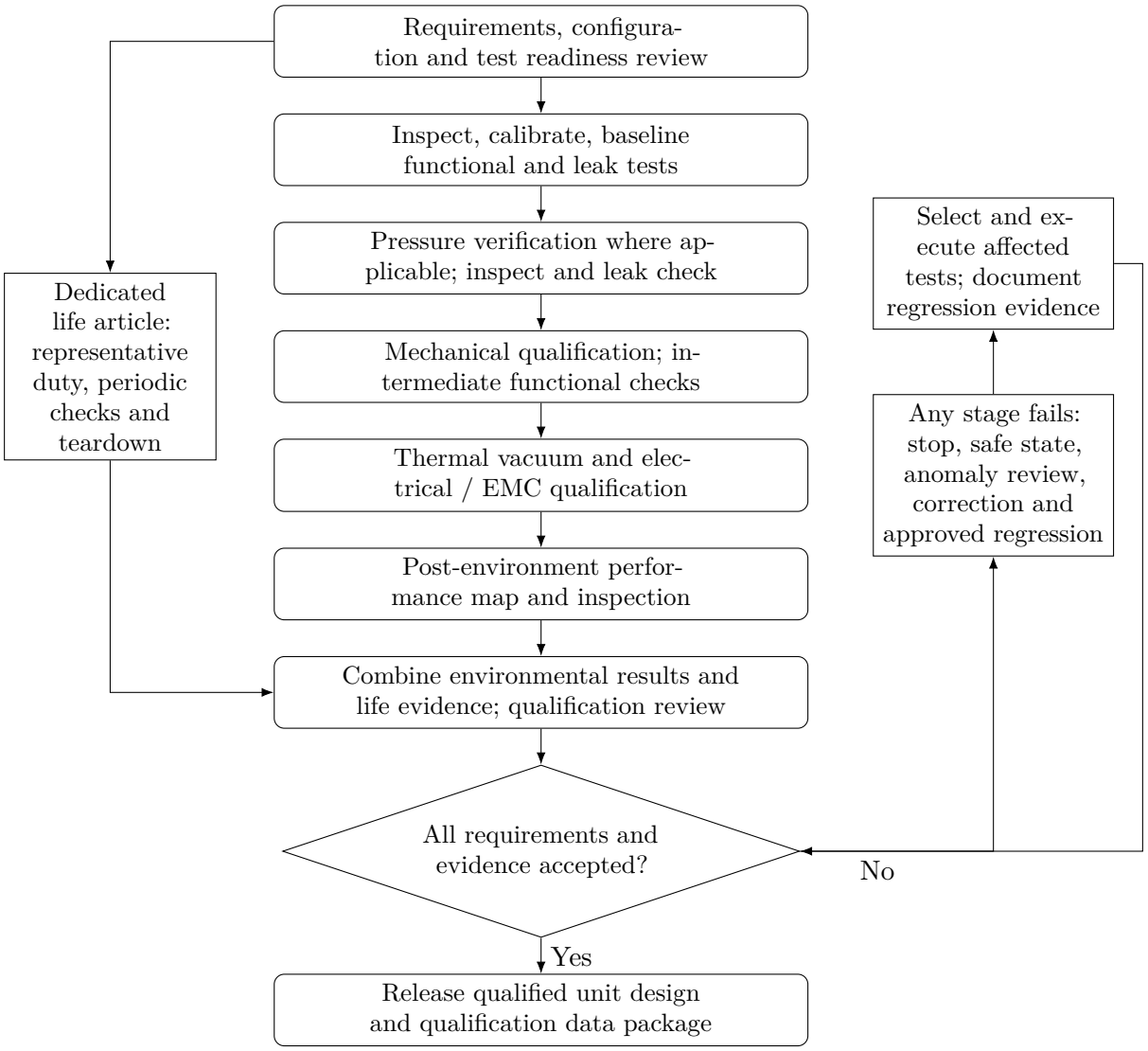
I plan a representative endurance string with continuous electrical, pressure, flow and temperature logging, reference-point checks every 100 operating hours and at the gas transition, and final grid/cathode metrology. Non-invasive checkpoints preserve the specified continuous segments; any intrusive inspections and restart exposure are recorded separately. I stop the test for loss of neutralisation, sustained current-ratio exceedance, an uncontained leak, temperature-limit exceedance or repeated arcs beyond the approved protection envelope. A change beyond the 5% performance-stability criterion triggers investigation and a formal disposition. Valve/ignition cycling is a separate exposure; its count is derived from the intended command profile rather than equating hours with starts. Select additional life allowance using degradation uncertainty, article variability and validated acceleration, rather than adopting an unexplained 1.5 or 2 multiplier. Accelerated sputter testing is valid only if energy, angle, surface chemistry and dominant mechanism remain representative. Valve and ignition cycles come from the commanded operational profile including contingencies. Until that profile exists, 500 starts or any other fixed count is a proposal.

A manufacturer total-impulse design target does not establish demonstrated qualification. Configuration, operating point, exposure history, failure criteria and test records are required for heritage credit. The assignment's 1000 h duty remains the primary requirement. Long-duration PPU operation can be supported by component stress/thermal analysis and appropriate endurance testing, while regulator and valve life needs pressure/cycle-specific evidence.

6.9 Test sequence and anomaly control

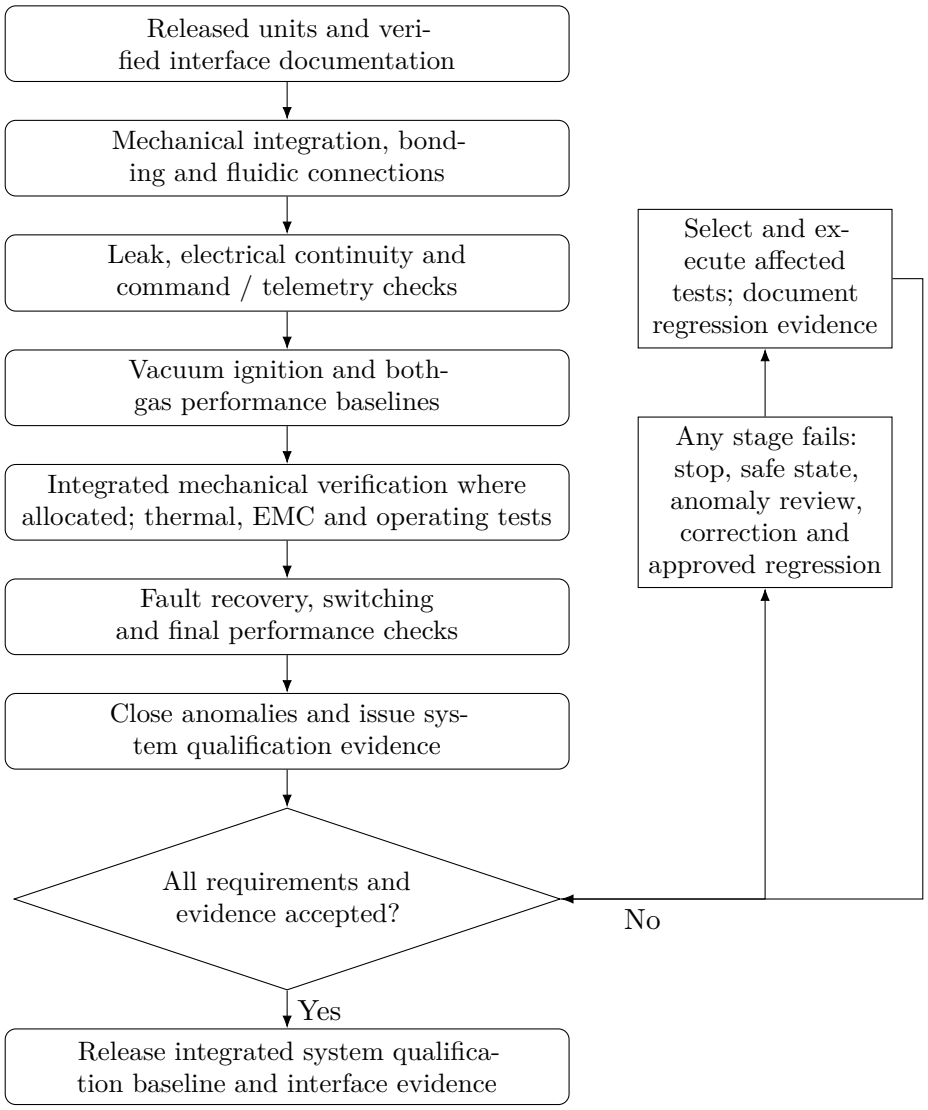
The following diagrams are proposed programme sequences. Pressure testing is performed by approved procedures before delicate firing operations where practical; exact order follows hazards, contamination controls and test objectives. Failure at any gate stops progression. Record the anomaly, preserve data, investigate root cause, approve corrective action and determine the affected regression scope. A repeat test alone does not close an unexplained failure. Qualification evidence must remain tied to hardware and software configuration.

Unit qualification flow



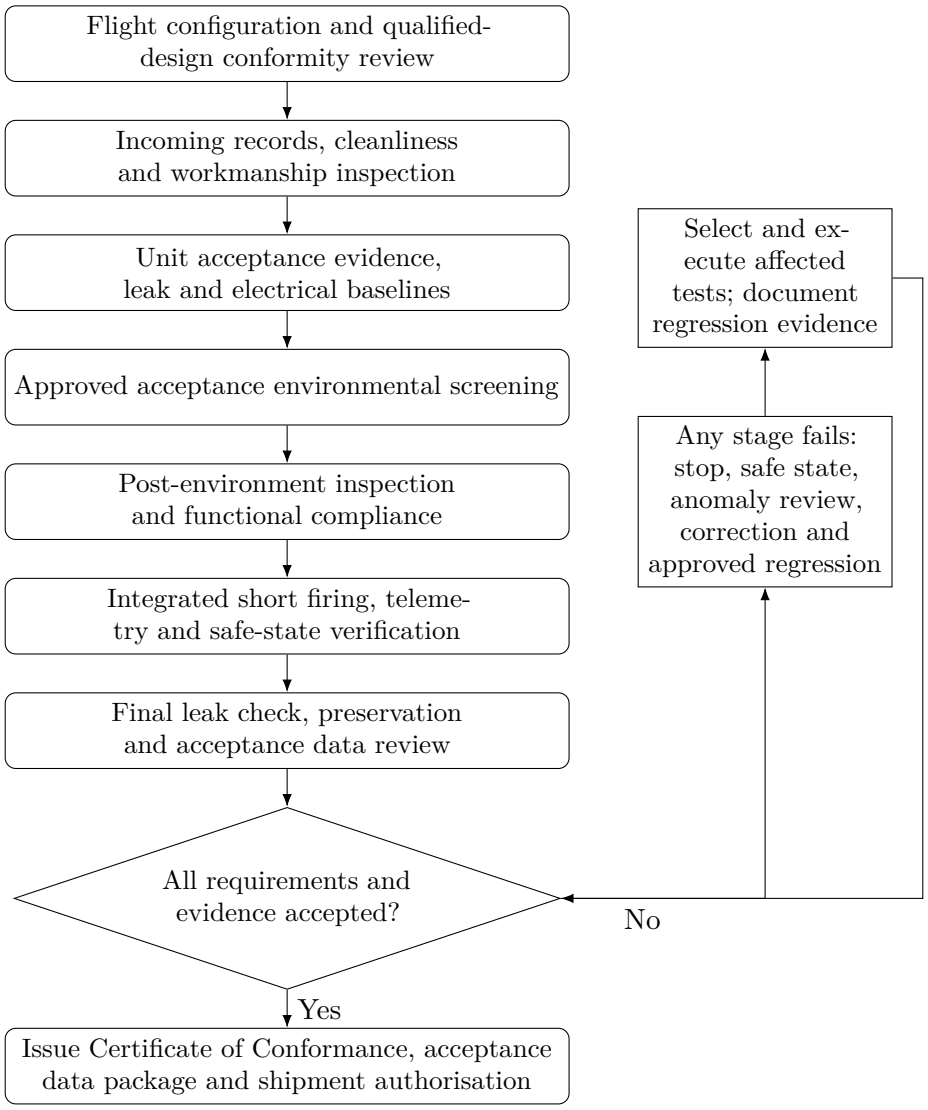
Each stage has a pass/fail gate. Any failure invokes the anomaly and regression process shown at right. The affected tests are selected from the failure impact assessment; evidence then returns to the release review. All remaining planned tests must also be complete.

Integration and system qualification flow



Each stage has a pass/fail gate. Any failure invokes the anomaly and regression process shown at right. The affected tests are selected from the failure impact assessment; evidence then returns to the release review. All remaining planned tests must also be complete.

Flight acceptance flow



Each stage has a pass/fail gate. Any failure invokes the anomaly and regression process shown at right. The affected tests are selected from the failure impact assessment; evidence then returns to the release review. All remaining planned tests must also be complete.

7 Design confirmations before implementation

Design item	Engineering significance	Confirmation evidence
Magnetic topology	Written axial-field request differs from illustrated reversal	Confirm centre-field solenoid interpretation or specify cusp reference points
Current topology and beam composition	Thrust interpretation and erosion model uncertain	Schematic, calibrated current accounting, species/angular diagnostics and thrust balance
Grid geometry and material	Perveance and local life cannot be calculated	Released apertures, cold/hot gap, surface state and temperature
Magnetic field and mounting	Centre-field design has 16.1% axial reduction at chamber ends	Required field uniformity, mounting clearances and measured field map
Coil thermal interface	Radiation-only temperature may underpredict heating	Mounted thermal test and correlated conductance/view factors
MFC definition and calibration	Throughput and specific impulse biased	Gas-specific calibration, standard reference and uncertainty
Tank pressure and oxygen service	Hot inventory may exceed allowable; ignition hazard	Certified pressure-temperature envelope, material and oxygen review
Neutraliser flow allocation	Tank mass and system performance understated	Clarify whether stated flows include cathode demand
Launch and spacecraft interfaces	Qualification spectrum and temperature assumptions invalid	Mounting-level loads, spectra, thermal boundary and EMC allocations
Life duty and configuration	Tests may not cover actual exposure	Starts, throughput, gas sequence, faults, validated life margin

8 Conclusions

Under the selected direct-interception assumption, the ideal estimates are 7.30/4.43 mN for xenon and 3.49/2.12 mN for the molecular mixture in Cases 1/2. The separately evaluated literature corrections reduce these to approximately 7.08/4.29 mN and 3.00/1.82 mN. The current interpretation and transferred correction factors are the principal uncertainties in this comparison. I would resolve them through current-path checks and direct thrust measurements before extended operation at the high-accelerator-current point.

The discrete 380-turn axial coil provides the required centre-field adjustment at 0.1125–1.125 A. Its nonuniformity and thermal interface remain part of the design definition. The molecular propellant requires a separate operating map, composition-aware interpretation and oxygen-compatible material verification.

Real-gas storage calculations confirm 59.60 bar xenon and 273.70 bar mixture at the selected loaded state. Xenon first condenses at approximately 13.37 °C at initial density. A conditional 300

bar hot limit requires reducing the mixture allowance or changing the thermal/hardware basis; it does not by itself prevent the nominal 500 h duty. The reduced inventory supports approximately 525 h before the assumed residual.

The verification programme defines unit, integrated and flight stages with measurable decision rules, baseline comparisons, anomaly handling and mission-derived endurance. The design confirmations in Section 7 provide a practical route from the calculated proposal to detailed implementation.

Additional calculations

For additional figures and graphs, refer to the accompanying `calculations.py`.

References

[1] A. Schwertheim et al., "Air-breathing electric propulsion developments at NewOrbit space," *Journal of Electric Propulsion*, 5, article 49, 2026. https://doi.org/10.1007/s44205-026-00226-5. Relevant sections: Hardware, Test plan, Experimental setup and Results.

[2] D. M. Goebel and I. Katz, *Fundamentals of Electric Propulsion: Ion and Hall Thrusters*, Wiley/JPL, 2008. https://doi.org/10.1002/9780470436448. Open author-hosted chapter collection: https://descanso.jpl.nasa.gov/SciTechBook/st_series1_chapter.html.

[3] J. Lopez-Uricoechea, D. Lev and M. L. R. Walker, "0-D composition and performance analysis of an air-breathing radiofrequency ion thruster," *Journal of Electric Propulsion*, 1, article 11, 2022. https://doi.org/10.1007/s44205-022-00013-y.

[4] CoolProp, *Mixtures*, version 8.0.0 documentation and HEOS property implementation. https://coolprop.org/fluid_properties/Mixtures. Accessed 31 October 2026.

[5] ECSS-E-ST-10-02C Rev.1, *Verification*, 1 February 2018. https://ecss.nl/standard/ecss-e-st-10-02c-rev-1-verification-1-february-2018/.

[6] ECSS-E-ST-10-03C Rev.1, *Testing*, 31 May 2022. Sections 4–5, equipment qualification and acceptance schedules. https://ecss.nl/home/ecss-e-st-10-03c-rev-1-testing-31-may-2022/.

[7] ECSS-E-ST-32-02C Rev.2, *Structural design and verification of pressurized hardware*, 15 October 2025. https://ecss.nl/wp-content/uploads/2025/10/ECSS-E-ST-32-02C-Rev.2(15October2025).pdf.

[8] ASTM G93/G93M-25, *Standard Guide for Cleanliness Levels and Cleaning Methods for Materials and Equipment Used in Oxygen-Enriched Environments*, 2025. https://doi.org/10.1520/G0093_G0093M-25. Cleanliness levels and implementation require project-specific selection.

[9] J. Zorn et al., *Characterization of the performance of a RIT-10 and rf-neutralizer for an ABEP system*, Journal of Electric Propulsion 4, 68 (2025). https://doi.org/10.1007/s44205-025-00170-w.

[10] NewOrbit Space, *Technology: NEO-1 and AURA*, manufacturer-published mission context and capability statements. https://neworbit.space/technology. Accessed 31 October 2026.

[11] NewOrbit Space, *Introducing the Latest Version of AURA*, 1 September 2026. https://neworbit.space/news/introducing-latest-version-of-aura.

[12] NewOrbit Space, *AURA Enters Preliminary Design Review*, 4 September 2026. https://neworbit.space/news/aura-enters-preliminary-design-review.

[13] *Langmuir Probe Distortions and Probe Compensation in an Inductively Coupled Plasma*, NASA Technical Reports Server record 20060056251. https://ntrs.nasa.gov/citations/20060056251.

[14] *Recommended practice for use of ExB probes in electric propulsion testing*, Journal of Electric Propulsion (2026). https://doi.org/10.1007/s44205-026-00225-6.

One-page summary of my previous experience

I have worked on both sides of electric propulsion development, measuring thrusters and simulating them. On the measurement side I ran the plume diagnostics for Hall thruster test campaigns and contributed to a flight diagnostic payload. On the modelling side I have done particle-in-cell work on Hall and RF thrusters, RF and helicon antenna coupling, and magnetic-circuit design in COMSOL and FEMM. Most of what I have learned has come from the points where the two disagreed.

Plasma and plume diagnostics – ISRO-LPSC / IIST, Junior Research Fellow. I developed the *Integrated Diagnostic Module* for ISRO's Technology Demonstration Satellite-01, a flight payload combining a quartz crystal microbalance, a retarding potential analyser and a Faraday probe for 300mn Hall thruster(estme.) and the CEX plasma diagnostics in orbit. I carried out the structural and thermal analyses for space qualification and launch compatibility, and contributed to the payload-to-bus data transmission architecture over the MIL-STD channel. On the ground I designed the experimental setups for laser-induced fluorescence and multispectral imaging, analysed the vacuum-chamber test data, and automated the in-chamber probe diagnostics on the facility's robotic positioning system; the acquisition sequence, sweep logic and calibration were mine. Multispectral imaging was used to estimate Hall thruster channel erosion and life, the same measurement problem as tracking grid erosion through a long-duration test. I also did the preliminary design of an Orbitrap mass spectrometer as a technology demonstrator. Owning that automation with my mechanical engineering background is why I am comfortable taking a measurement chain end to end, from probe geometry and bias scheme through calibration to the uncertainty on the published number.

Test campaigns and procedures. I ran the plume diagnostics on the Hall thruster test campaigns, which meant owning the measurement from probe setup and bias scheme through acquisition to the reduced result, and reporting it in a form the design side could act on. I also supported the test and evaluation campaign for the Power Processing and Control Unit, which failed repeatedly in ground testing before the cause was found and resolved. I was not leading that campaign, but being on it is where I saw how differently an integrated chain can behave from its parts on the bench. That is the reasoning behind the qualification and flight-acceptance structure in Section 6, and behind treating the oxygen-compatibility and cleanliness provisions in Section 5 as gates rather than attributes of incoming inspection.

Simulation-based design and radio-frequency modelling. At ISRO I analysed xenon Hall-effect *and radio-frequency* thruster performance in VSim, ANSYS and MATLAB, including particle-in-cell simulation with ceramic erosion from ion impact, and ran helicon and RF antenna coupling simulations for wave-to-plasma power transfer and ionisation, which sets the matching behaviour of an RF thruster and an RF cathode. For my semester thesis at the TUM Chair of Space Mobility and Propulsion, with Dr Henrike Jakob, I am developing a miniaturised water-fed coaxial pulsed plasma thruster, running a 2D axisymmetric PIC-DSMC simulation of the vaporisation dynamics, optimising the PPU design, and building the diagnostic concept around RPA, Faraday probes and optical plume characterisation. I also designed a sub-kilowatt water-fed Hall thruster concept targeting 14.75 mN at 1500 s, using COMSOL and FEMM for the magnetic circuit, electron confinement and ionisation region, the same class of problem as Section 3 here.

Plasma physics foundation. At the Max Planck Institute for Plasma Physics in Garching I attended the summer school on stellarator theory and optimisation, MHD, kinetic theory and plasma diagnostics, then worked with Dr Gregor Birkenmeier on magnetic reconnection in tokamak sawtooth instability, using 2D MHD codes that resolve thin current sheets and capture plasmoid instabilities in the Sweet–Parker model. At TUM I have taken Plasma Physics 1 and 2 and the electric propulsion course.

Tools, output and languages. **Software:** Python, C++, C, MATLAB/Simulink, run in parallel where needed; VSim, COMSOL, FEMM, ANSYS, LTspice, LabVIEW, SolidWorks, Siemens NX; ROS for the probe automation. **Instrumentation:** experimental setups for multispectral imaging, laser-induced fluorescence and Orbitrap mass spectrometry. **Other details:** first-author paper in a Springer journal (2025); a poster on digital-twin formation control for Hall-thruster constellations selected for i-SAIRAS/iSpaRo 2026, Cologne; a paper at GLEX 2025; an Indian patent (545629); attendance at the 1st German Electric Propulsion Conference, Giessen. **Languages:** English C1, German A2 and improving, Hindi C1, Marathi native.

Plan for the first 1, 3, and 6 months

Programme priorities and scope

The immediate priority would be reliable diagnostic and qualification support for the xenon propulsion system. In our discussion, Ruslan identified the first-flight xenon variant as the priority, described Faraday-probe commissioning and existing RF-compensated plasma measurements, and outlined parallel Hall-thruster development. These discussion points guide the proposed work; the commissioning status and prototype schedule would be confirmed on joining.

The public programme provides a useful starting point. NewOrbit announced six EM2 units and an expected qualification completion by the end of Q1 2027, followed by flight preparation. Its September 2026 update reported entry into Preliminary Design Review [11, 12]. The June 2027 flight-model target discussed during the interview is treated as a separate delivery milestone, not a replacement for the published qualification target. Months below run from the actual start date and would be aligned with the team's current test schedule.

The plan builds on existing hardware, procedures and models. Diagnostic changes would first be demonstrated on development equipment and introduced into qualification only through configuration control. In-house probe development is the starting point; specialised external facilities would be used where they provide necessary capability or independent verification without delaying the flight priority.

By the end of month 1: establish repeatable measurements

The first weeks would combine facility familiarisation with a bounded measurement task. This includes the vacuum, high-voltage, RF and gas-system procedures, existing interlocks, instrumentation calibrations and test configuration records. The Faraday probe and traverse are a practical starting point if commissioning is still open; otherwise, the priority becomes repeatability and uncertainty assessment of the commissioned system.

The commissioning procedure would cover collector and guard geometry, material and surface condition, bias sweeps, secondary-electron effects, alignment, position repeatability and probe distance. A divergence result needs an explicit definition, such as an agreed enclosed-current half-angle, rather than an unspecified angle. Repeated scans at reference operating points would establish repeatability and sensitivity to bias, distance and background pressure. Any conversion from angular charge current to momentum divergence would state its species and energy assumptions.

A selected subset of the published xenon performance map would be reconstructed from available internal raw data, including current definitions, flow reference conditions, balance calibration, tare and data rejection. The deliverable would be a documented reduction pipeline and an uncertainty budget, with discrepancies resolved against the original configuration rather than assuming that EM2 must reproduce its predecessor exactly.

Facility comparison would use measured operating pressure, gas correction, pumping speed, conductance, gas load and gauge position. Chamber volume affects pumpdown and transients but does not by itself determine steady operating pressure. Every performance point would retain its facility configuration and background conditions.

Month 1 outcome: an agreed diagnostic procedure, repeatability results for selected xenon operating points, a traceable data-reduction check and an initial thrust and specific-impulse uncertainty budget. Full commissioning remains conditional on hardware readiness and test access.

By the end of month 3: resolve the important diagnostic uncertainties

The existing Langmuir-probe compensation would be evaluated at the actual RF frequencies, operating points and cable configuration. Residual RF at the probe, compensation impedance, sweep-rate sensitivity, leakage and repeatability would be checked before interpreting electron temperature or density. Comparison with a characterised reference arrangement would establish whether a change is needed. The objective is to quantify validity and operating limits, not to presume that the present engineering approach is inadequate. Probe perturbation, magnetic-field effects and the assumed electron distribution remain part of the interpretation [13].

With a repeatable angular scan available, an RPA could add energy-per-charge information where this would resolve a specific performance or erosion question. It cannot, by itself, separate charge states: ions accelerated through the same potential can have the same energy per charge. Charge-state assessment requires an additional diagnostic, such as an ExB velocity filter with a justified acceleration model, together with transmission, energy-spread and angular corrections [14]. Consequently, the deliverable would be a measured divergence contribution and a justified residual correction or species estimate, not an unconditional promise to decompose the entire thrust factor.

Thruster-only and system performance would be reported with explicit boundaries. At fixed thrust, including cathode flow gives

$$I_{sp,system} = I_{sp,thruster} \frac{\dot{m}_t}{\dot{m}_t + \dot{m}_c}. \tag{15}$$

This is a bookkeeping correction for the same measured operating point, not a prediction that changing cathode flow leaves thrust unchanged. System efficiency would additionally include PPU, RF-generator, neutraliser and auxiliary power.

Air-mixture species measurements remain a secondary development task unless they directly support the flight campaign. An ExB measurement would require care with overlapping mass-to-charge ratios, energy distributions and angular sampling; a single on-axis trace cannot establish the entire beam composition.

Month 3 outcome: a quantified assessment of RF-compensation performance, a validated diagnostic operating procedure, improved xenon plume characterisation and consistently defined system-level performance. Additional probe construction would proceed only where the uncertainty or design decision justifies it.

By the end of month 6: support qualification and endurance decisions

The main responsibility would be an agreed measurement work package within the qualification campaign, coordinated with the propulsion, electronics and AIT leads. Continuous records would include grid currents and voltages, RF forward/reflected power, gas flows, pressure, temperatures, neutraliser coupling and protection events. Reference-point thrust and angular scans would be repeated at planned checkpoints and following environmental exposure. Gas switching, ignition delay, recovery and repeatability would be assessed against the approved operating sequence.

Accelerator-to-screen current ratio is a useful health indicator, but the published 4% data-selection threshold would not automatically become a universal acceptance limit. Trends would be interpreted alongside thermal state, facility pressure, voltage, alignment and the qualified operating envelope. Checkpoints would be planned to preserve continuous endurance segments where required, with interruptions and restart exposure recorded explicitly.

Wear assessment would combine pre/post aperture and gap metrology, backstreaming-margin measurements and trends in performance and grid current. A QCM could provide deposition mon-

itoring after accounting for temperature, viewing geometry, sticking and re-sputtering; deposited mass is not a direct measurement of grid erosion. Optical emission could support investigation of material release where suitable lines, sensitivity and background discrimination are established. Neither technique would replace dimensional wear measurements.

A bounded model-comparison task would connect measured operating conditions to aperture growth or another agreed degradation observable. Validation would require comparison at independent conditions and uncertainty propagation before using the model for life extrapolation. Accumulated time, throughput, starts, materials degradation and electronic reliability all contribute to the life case; one erosion-rate measurement cannot establish complete system life.

Hall-thruster work would follow the flight-system priority and actual prototype availability. Probe concepts may be reusable, but voltage range, heat loading, bandwidth, magnetic sensitivity, plume geometry and calibration must be checked for the new device. The first contribution would be a diagnostic requirements and commissioning plan, followed by baseline measurements when the prototype is ready. No Hall-thruster commissioning date is assumed from the public programme.

Month 6 outcome: a traceable diagnostic data package supporting the assigned qualification/endurance activities, documented degradation trends and a tested model comparison where sufficient wear is measurable. Hall-thruster diagnostic support would be delivered to the maturity available, without committing to complete qualification or life validation within six months.

Assumptions and basis for revision

Certain design and test-planning assumptions in this submission are my own engineering choices, informed by the assignment, NewOrbit's published research and mission information, and the programme priorities discussed during the interview. I have checked the calculations and supporting sources as carefully as possible against the information available on 31 October 2026. Public capability statements, programme targets and proposed test criteria are distinguished from demonstrated results. The assumptions and work plan can be revised when the agreed start date, current hardware configuration, internal test evidence and mission requirements are available.

My MSc semester thesis fell due in the same window as this assignment, so I had less room for back-and-forth than I would have liked with the NewOrbit Space team. Where a point would normally have been a short question to the team, I resolved it myself and wrote the reasoning down instead. The places where that happened are flagged in the text: the two current interpretations in Task 1, the conditional 300 bar case in Task 4, and the limits of what the published vibration evidence supports in Task 5. If any of those readings differs from what was intended, the affected result can be re-derived quickly, because each one is isolated to a stated assumption.

I have also prepared a slide version of the same material. It covers the five tasks in order, the thrust and specific impulse derivation with both current models, the coil design and field results, the propellant transition, the storage calculation and the qualification and acceptance structure, followed by the 1, 3 and 6 month plan. It is meant as something to talk through during the review and not as a second document to read; everything in it traces back to this report.

Every number in both comes from the supplied `calculations.py`. The storage results have since been reproduced on a separate CoolProp 8.0.0 installation. A Peng–Robinson cross-check agrees with the reference equation to 0.8% for xenon but differs by 3.5% for the mixture, which is about 9 bar at the 20 °C loaded state. The reference equation is used throughout and the cubic equation only as a check, so that spread should be carried as an analysis uncertainty until the tank supplier's own property basis is agreed. Refer this git repo for more details: `https://github.com/omkulkarni01/Assignment_NewOrbit_space.git`